



Experimental AI Subseasonal to Seasonal (S2S) Forecast Guidance for India

Forecast Initialization: 19 August 2026

Forecast Period: 19 August - 15 September 2026

Forecast Range: Weeks 1-4

Reference Dataset: ERA5 Reanalysis / model climatology, 2002-2021

Forecast Product: 100-member ensemble mean

Forecast Initialization: IFS, GFS and delayed ERA5

Status: Experimental Research Guidance

Executive Summary

For the 19 August 2026 initialization, the S2S guidance depicts a monsoon pattern evolving from a central-India-focused rainfall regime during Weeks 1-2 toward an increasingly eastern and northeastern rainfall regime during Weeks 3-4. The first week favors enhanced rainfall mainly over eastern Madhya Pradesh and adjoining central India, while much of the rest of the country remains comparatively dry. During Week 2, a possible low-pressure circulation may sustain rainfall over eastern Madhya Pradesh, although Arabian Sea southwesterlies remain weak. By Weeks 3 and 4, positive rainfall anomalies migrate eastward and become concentrated over eastern and northeastern India. Simultaneously, anomalous easterlies strengthen over the southern peninsula and Arabian Sea, rainfall decreases over western and central India, and positive temperature anomalies become increasingly prominent across northern and northwestern India. Taken together, the rainfall, temperature and circulation fields suggest a progressive weakening and eastward reorganization of the southwest monsoon circulation from late August into mid-September, with an emerging signal consistent with the beginning of the seasonal transition toward monsoon withdrawal.

Forecast details including results with initialisations from IFS, GFS and delayed ERA5 are available at <https://s2s.scdlds.com/>

1. Methodology

This report summarizes the experimental subseasonal-to-seasonal (S2S) forecast guidance for India initialized on 19 August 2026, covering the four-week period from 19 August to 15 September 2026. Below, the SCDLDS forecast product uses an IFS operational initialization and a 100-member ensemble mean for Weeks 1-4. See <https://s2s.scdlds.com/> for ERA 5 delayed and GFS initialisation forecasts. The assessment focuses on the evolution of weekly



rainfall, rainfall anomalies, 2-m air temperature anomalies, and 850-hPa circulation anomalies. Rainfall and temperature departures are evaluated relative to the model's typical conditions for the corresponding season and forecast lead during 2002-2021. The forecast indicates a progressive eastward displacement of anomalously wet conditions during the forecast period. In Week 1, the strongest positive rainfall anomaly is concentrated over central India, particularly toward eastern Madhya Pradesh and adjoining areas, whereas much of the remainder of the country experiences near-normal to below-normal rainfall. During Week 2, enhanced rainfall remains focused over central/eastern India, consistent with the possibility of a low-pressure system influencing eastern Madhya Pradesh and nearby regions. By Weeks 3-4, the positive rainfall signal shifts farther east, becoming increasingly concentrated over eastern and northeastern India, while western, central, and parts of peninsular India become comparatively dry. The later forecast weeks also show increasing positive temperature anomalies across northern, northwestern and central India. The 850-hPa circulation evolution provides additional dynamical support for this transition. The early forecast weeks indicate relatively weak anomalous monsoon westerlies, while Weeks 3-4 show increasingly prominent anomalous easterly flow over the southern peninsula and adjoining Arabian Sea. This circulation evolution is consistent with a weakening of the large-scale southwest monsoon circulation and may represent an early indication of the seasonal withdrawal phase, although the timing of monsoon withdrawal should not be inferred from a single experimental forecast initialization alone.

2. Rainfall Forecast

2.1 Week 1: 19-25 August 2026: The Week 1 rainfall anomaly forecast shows a pronounced regional contrast across India. The principal positive rainfall anomaly is concentrated over central India, particularly eastern Madhya Pradesh and adjoining portions of central and east-central India. This region stands out against predominantly weak negative rainfall anomalies across much of western India, the western peninsula and several northern and eastern regions. The absolute rainfall forecast also indicates substantial rainfall over the central monsoon zone and eastern portions of the country during this period. The uploaded S2S product defines Week 1 as 19-25 August 2026. The distribution suggests that the active component of the monsoon circulation is relatively localized rather than widespread across the country. Rainfall activity over parts of the western monsoon region and southern peninsula is comparatively suppressed. The corresponding 850-hPa circulation anomaly indicates a weakened monsoon flow over the Arabian Sea and peninsular India. The anomalous circulation is also consistent with an east-west shear zone displaced relatively northward, limiting the organization of widespread monsoon rainfall over much of the country.

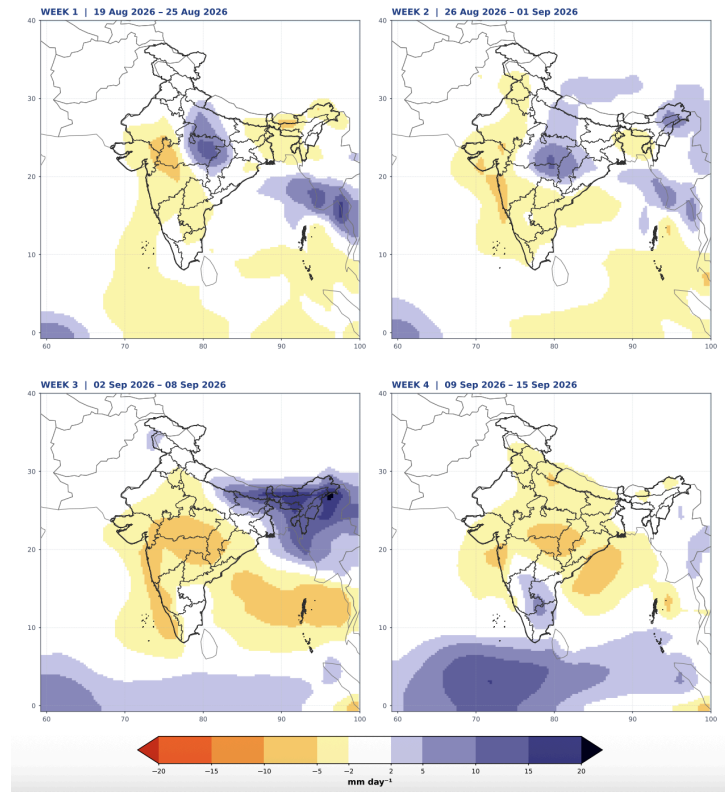


Figure 1. Weekly mean rainfall anomaly over India for Weeks 1-4, IFS initialized 19 August 2026. Panels: (a) 19-25 August, (b) 26 August-01 September, (c) 02-08 September, and (d) 09-15 September 2026. Anomalies represent departures from the model's typical rainfall for the corresponding season and forecast lead during 2002-2021.

2.2 Week 2: 26 August-01 September 2026: During Week 2, a coherent positive rainfall anomaly remains evident across central to east-central India, with the strongest signal extending across parts of Madhya Pradesh and adjoining regions. The spatial organization is consistent with the possibility of a low-pressure system or cyclonic circulation progressing inland toward eastern Madhya Pradesh, thereby sustaining enhanced rainfall over this sector. Outside this region, much of western India and substantial portions of the peninsula continue to exhibit negative rainfall anomalies. The anomalous southwesterly flow over the Arabian Sea also remains relatively weak, suggesting that the broader monsoon westerly circulation is not particularly strong during this period. The Week 2 forecast period extends from 26 August to 01 September 2026. The rainfall distribution therefore continues to favor a regionally concentrated monsoon response, rather than a spatially extensive wet spell across India.

2.3 Week 3: 02-08 September 2026: A substantial reorganization of rainfall anomalies emerges during Week 3. Positive rainfall anomalies shift markedly eastward and become concentrated across the eastern Indo-Gangetic region, eastern India and northeastern India. In contrast,



broad negative rainfall anomalies expand across much of western and central India, including portions of Maharashtra, Madhya Pradesh and adjoining areas. The strongest anomalously wet region during this period lies considerably farther east than in Weeks 1-2, highlighting a progressive eastward displacement of the active rainfall anomaly. The Week 3 rainfall map therefore suggests a weakening of rainfall activity over the western and central monsoon regions while eastern and northeastern India remain comparatively active. The circulation forecast provides an important additional signal. At 850 hPa, anomalous easterly flow becomes more evident across the southern peninsula and adjoining tropical Indian Ocean/Arabian Sea sector, while the climatological monsoon southwesterly circulation appears increasingly suppressed. This combination of suppressed rainfall over western and central India and anomalous low-level easterlies over southern India may indicate the beginning of a broader transition in the seasonal monsoon circulation.

2.4 Week 4: 09-15 September 2026: By Week 4, much of western, central and northern India is characterized by negative rainfall anomalies, whereas the comparatively favorable rainfall signal is displaced toward the far eastern and northeastern part of the country. The most prominent dry anomalies extend across parts of central India and adjoining north-central regions, while enhanced rainfall becomes much less widespread over the core monsoon zone. Some positive rainfall anomalies remain over parts of southern peninsular India and adjacent oceanic regions, but the large-scale continental rainfall pattern is dominated by suppressed conditions across the central and western sectors. The Week 4 forecast covers 09-15 September 2026. The west-to-east progression evident through the four weeks, from a central Indian rainfall enhancement during Weeks 1-2 to an increasingly eastern/northeastern focus thereafter represents one of the clearest subseasonal signals in this forecast initialization.

3. Wind anomaly forecasts

The 850-hPa wind anomaly forecasts provide dynamical context for the evolution of rainfall and temperature over the four-week forecast period. The maps show weekly wind-speed anomalies together with anomalous U/V wind vectors relative to the 2002-2021 reference period.

Week 1: During Week 1, the low-level circulation indicates relatively weak anomalous monsoon westerlies over the Arabian Sea and peninsular India. The anomalous circulation suggests that the main monsoon shear zone is positioned relatively northward, closer to the Indo-Gangetic and Himalayan foothill region, while the conventional southwesterly flow into the Indian landmass is weaker than normal. This circulation configuration is consistent with enhanced rainfall being concentrated over central/east-central India rather than being distributed uniformly across the monsoon zone.

Week 2: During Week 2, anomalous circulation over central India remains favorable for maintaining rainfall around eastern Madhya Pradesh and adjoining regions. However, the southwesterly monsoon circulation over the Arabian Sea remains weaker than climatology.

Thus, despite the possibility of a regional low-pressure circulation over central India, the broader southwest monsoon flow remains subdued.

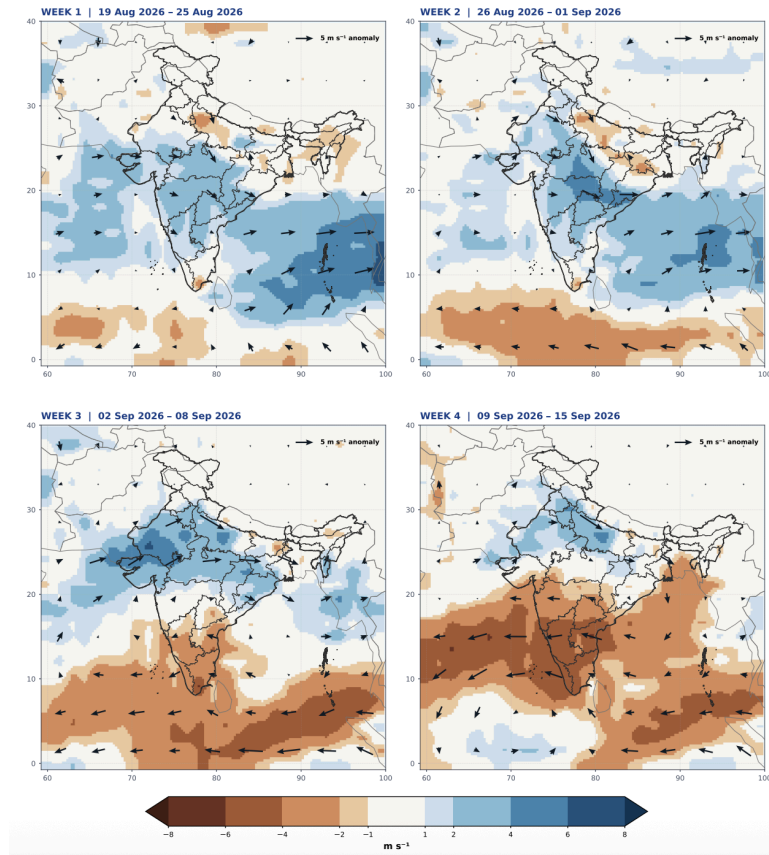


Figure 3. Weekly mean wind anomaly over India for Weeks 1-4, IFS initialized 19 August 2026. Anomalies are relative to the model's typical temperature for the corresponding season and forecast lead during 2002-2021.

Week 3: By Week 3, the circulation pattern undergoes a pronounced transition. Anomalous easterlies become evident over the southern peninsula and adjoining Arabian Sea, while anomalous circulation farther north remains comparatively favorable for rainfall over eastern and northeastern India. This circulation change occurs simultaneously with the development of negative rainfall anomalies across western and central India, providing a dynamically consistent indication of weakening monsoon conditions over these regions.

Week 4: The transition becomes still more pronounced during Week 4. Strong anomalous easterly flow dominates much of the southern peninsula, Arabian Sea and adjoining central Indian Ocean sector, while anomalous westerly monsoon flow is substantially weakened. At the same time, negative rainfall anomalies cover much of central and northern India, and positive temperature anomalies become widespread. Taken together, these features are consistent with

a progressive weakening and reorganization of the southwest monsoon circulation. They may indicate the early stages of the seasonal transition toward monsoon withdrawal, although actual withdrawal should be assessed using established operational criteria and subsequent forecast initializations rather than this S2S product alone.

4. Temperature Forecast

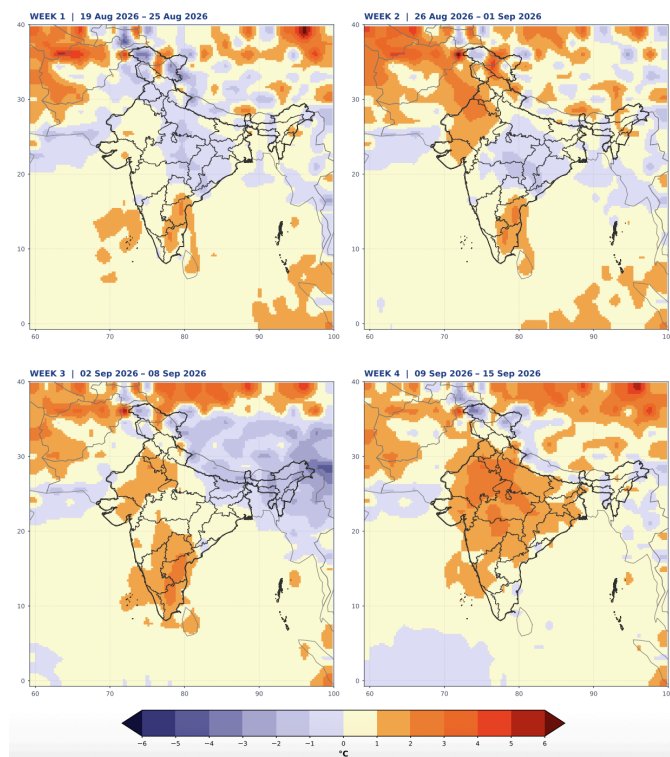


Figure 2. Weekly mean 2-m temperature anomaly over India for Weeks 1-4, IFS initialized 19 August 2026. Anomalies are relative to the model's typical temperature for the corresponding season and forecast lead during 2002-2021.

4.1 Week 1: 19-25 August 2026: During Week 1, below-normal temperature anomalies dominate substantial portions of central, east-central and northern India. These relatively cool conditions broadly coincide with the region of enhanced rainfall over central India and are physically consistent with increased cloud cover, rainfall and reduced daytime surface heating. Warmer-than-normal conditions are more evident over localized parts of southern peninsular India and regions farther to the northwest, while temperature departures over much of the remainder of India are relatively modest. Overall, the Week 1 temperature pattern reinforces the rainfall signal, with anomalously wet central regions generally experiencing neutral to negative temperature anomalies.

4.2 Week 2: 26 August-01 September 2026: During Week 2, negative temperature anomalies persist over a broad swath of central and eastern India, once again corresponding approximately to the region where rainfall remains enhanced. At the same time, positive temperature anomalies become more evident across northwestern India, particularly toward Rajasthan and adjoining regions. This northwestern warming is consistent with the comparatively dry conditions forecast over this sector and the absence of strong anomalous monsoon rainfall. Parts of southern peninsular India also remain warmer than normal, while eastern and central regions are comparatively cooler. The forecast therefore begins to show a more pronounced temperature contrast between the drier northwest and wetter central/eastern sector.

4.3 Week 3: 02-08 September 2026: A clear change in the spatial temperature pattern develops during Week 3. Positive temperature anomalies expand across western, central and southern peninsular India, coinciding with the development of widespread negative rainfall anomalies across these regions. Meanwhile, temperatures across parts of the eastern Indo-Gangetic plain, eastern India and Northeast India remain close to normal or below normal, consistent with the eastward displacement of enhanced rainfall. This configuration produces a distinct east-west contrast: relatively warm and dry conditions dominate western and central India, while the eastern sector remains comparatively wetter and cooler.

4.4 Week 4: 09-15 September 2026: By Week 4, positive temperature anomalies strengthen substantially across northwestern, northern, central and parts of western India. The warming is spatially coherent across much of the region experiencing suppressed rainfall, particularly over central and northwestern India. In several locations, the ensemble-mean temperature anomaly reaches approximately $+2^{\circ}\text{C}$ to $+4^{\circ}\text{C}$, although local values vary spatially. Eastern and northeastern India remain comparatively less warm, broadly consistent with relatively greater rainfall activity in the eastern sector. The combination of a broad dry rainfall anomaly and enhanced positive temperature anomaly over northern and northwestern India represents a notable late-period signal and is consistent with increased surface heating under reduced cloudiness and precipitation.

5. Key S2S Signals

- **Central Indian rainfall enhancement during Weeks 1-2:** The strongest positive rainfall anomalies during the first half of the forecast period are concentrated over central to east-central India, with eastern Madhya Pradesh and adjoining areas forming an important focus.
- **Possibility of a central Indian low during Week 2:** The Week 2 rainfall and circulation configuration is consistent with the potential development or passage of a low-pressure circulation toward eastern Madhya Pradesh, supporting locally enhanced rainfall.
- **Predominantly dry conditions elsewhere:** Large portions of western, northwestern and peninsular India remain near normal to drier than normal during much of the forecast period.

- **Eastward shift of enhanced rainfall during Weeks 3-4:** The positive rainfall anomaly migrates toward eastern and northeastern India during the later forecast weeks, while rainfall becomes increasingly suppressed across western and central India.
- **Weak Arabian Sea monsoon westerlies:** The low-level circulation suggests weakened anomalous southwesterly flow over the Arabian Sea during the early forecast weeks.
- **Increasing anomalous easterlies during Weeks 3-4:** Easterly anomalies strengthen over the southern peninsula and adjoining Arabian Sea during the latter half of the forecast, indicating a weakening or reorganization of the southwest monsoon circulation.
- **Progressive warming over northern and northwestern India:** Positive temperature anomalies become increasingly widespread from Week 2 onward and are particularly prominent across northwestern, northern and central India by Week 4.
- **Rainfall-temperature consistency:** Regions experiencing suppressed rainfall generally develop positive temperature anomalies, while areas retaining enhanced rainfall, particularly toward eastern India, tend to remain comparatively cooler.
- **Possible early withdrawal signal:** The combination of suppressed rainfall over western/central India, anomalous easterlies over southern India and the Arabian Sea, and increasing positive temperature anomalies over northern and northwestern India is consistent with the early stages of a seasonal monsoon transition.

6. Interpretation

The 19 August 2026 initialization suggests a meaningful subseasonal transition in the spatial organization of the Indian summer monsoon during the period extending from late August into mid-September. During Weeks 1-2, rainfall activity is preferentially concentrated over central and east-central India, particularly in and around eastern Madhya Pradesh. The circulation pattern suggests that this enhancement occurs despite relatively weak large-scale southwesterly monsoon flow over the Arabian Sea, indicating that regional synoptic-scale disturbances may play an important role in organizing rainfall during this period. During Weeks 3-4, the rainfall pattern undergoes a marked eastward displacement. Enhanced rainfall becomes progressively concentrated toward eastern and northeastern India, while large portions of western, central and eventually northern India experience below-normal rainfall. The temperature response evolves consistently with this rainfall redistribution. Initially, cooler conditions accompanied enhanced rainfall over central India. Subsequently, as rainfall becomes suppressed over western, central and northwestern India, positive temperature anomalies expand and intensify across those regions. Perhaps the most noteworthy large-scale circulation signal is the development of anomalous easterlies over the southern peninsula and Arabian Sea during Weeks 3-4. Together with weakening rainfall over the central and western monsoon regions, this suggests a substantial weakening of the mature southwest monsoon circulation. The evolution is therefore consistent with a possible early transition toward southwest monsoon withdrawal, particularly during the first half of September. This signal should, however, be interpreted cautiously. S2S forecasts at Weeks 3-4 contain substantially greater uncertainty than shorter-lead forecasts, and formal monsoon withdrawal requires consideration of operational



meteorological criteria beyond rainfall and 850-hPa wind anomalies alone. Successive forecast initializations should therefore be monitored to determine whether the projected eastward migration of rainfall, weakening Arabian Sea monsoon flow, development of anomalous easterlies, and increasing warmth over northwestern India remain persistent features of the forecast.